Grade 9 – Mathematics

Third Term Model Paper 1

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Name / Index No.: ..... **Time: 3 hours**

- This paper consists of two parts: Part I and Part II.
- Part I: Answer **all** questions on this paper itself. Part A has 15 questions (2 marks each). Part B has 2 questions (10 marks each).
- Part II: Answer **five** questions only, on separate sheets. Each question carries 10 marks.
- Show all relevant steps of working. Write units where necessary. Use π = 22/7 where needed.

For examiner's use only	
PART I – A	/ 30
PART I – B	/ 20
PART II	/ 50
Total	**/ 100**

PART I

Part A – Answer all questions in the space provided. (2 marks each)

1. Write the general term (n^{th} term) of the number pattern 5, 8, 11, 14, … (2)

Answer: .....

2. Write the number 45 as a binary number (in base two). (2)

Answer: .....

3. Simplify: 2⅓ ÷ 1¾ (2)

Answer: .....

4. The marked price of a shirt is Rs 2400. Find its selling price when a 15% discount is given. (2)

Answer: .....

5. Factorise: $x^2 + 7x + 12$ (2)

Answer: .....

6. Simplify and give the answer with a positive index: $a^3 \times a^5 \div a^{-2}$ (2)

Answer: .....

7. Write 0.000507 in scientific notation. (2)

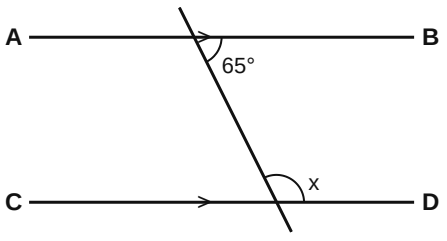
Answer: .....

8. Solve: $2(x - 3) = x + 4$ (2)

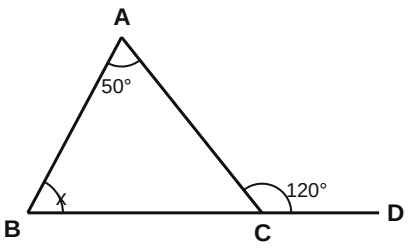
Answer: .....

9. In the figure, AB ∥ CD. Find the value of x. (2)

Answer: .....



10. In the figure, the side BC of triangle ABC is produced to D. Find the value of x. (2)

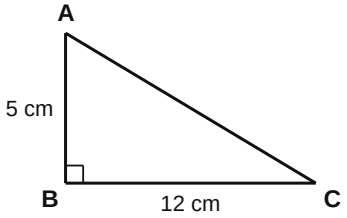


Answer: .....

11. Find the circumference of a circle of radius 7 cm. (π = 22/7) (2)

Answer: .....

12. Find the length of AC in the right-angled triangle ABC. (2)



Answer: .....

13. Solve the inequality $2x + 3 < 11$ and write the greatest integer value of x that satisfies it. (2)

Answer: .....

14. A = {factors of 12} and B = {factors of 18}. Write the elements of $A \cap B$. (2)

Answer: .....

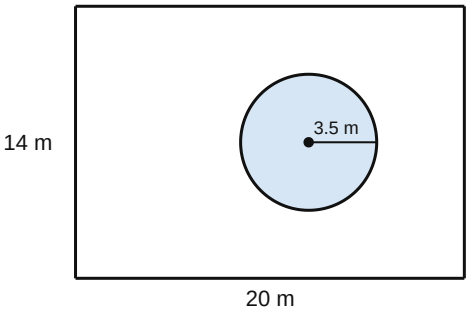
15. A bag contains 3 red balls and 5 blue balls of the same size. One ball is taken at random. Find the probability that it is red. (2)

Answer: .....

Part B – Answer both questions on this paper. (10 marks each)

1.

The figure shows a rectangular garden of length 20 m and breadth 14 m. There is a circular pond of radius 3.5 m in it. (π = 22/7)



(i) Find the area of the garden. (2 marks)

(ii) Find the area of the pond. (2 marks)

(iii) Find the area of the garden excluding the pond. (2 marks)

(iv) Find the circumference of the pond. (2 marks)

(v) A fence is built around the pond at Rs 150 per metre. Find the cost. (2 marks)

2.

In a class of 40 students, 25 play cricket, 18 play volleyball and 7 play both games.

(i) Represent this information in a Venn diagram. (Take C = {cricket players}, V = {volleyball players}.) (3 marks)

(ii) How many students play neither game? (2 marks)

(iii) How many students play cricket only? (2 marks)

(iv) A student is selected at random. Find the probability that he plays volleyball only. (3 marks)

PART II

Answer five questions only. (10 marks each)

1.

(a) A trader buys a television for Rs 48 000 and marks its price 25% above the cost price. He sells it giving a 10% discount on the marked price.

(i) Find the marked price. (2 marks)

(ii) Find the selling price. (2 marks)

(iii) Find the profit as a percentage of the cost price. (3 marks)

(b) The price of 5 kg of rice is Rs 1150. Assuming that the price is directly proportional to the amount, find the price of 8 kg of rice. (3 marks)

2.

(a) Factorise: (i) $2x^2 - 8$ (ii) $x^2 - 5x - 14$ (iii) $ax + ay + bx + by$ (5 marks)

(b) Simplify: $x/3 + (x + 1)/4$ (3 marks)

(c) Expand: $(2x - 3)^2$ (2 marks)

3.

(a) Solve: $(x + 2)/3 - (x - 1)/4 = 2$ (3 marks)

(b) The total price of 2 pens and 3 books is Rs 340. The total price of 4 pens and 1 book is Rs 280. Taking the price of a pen as Rs p and the price of a book as Rs b, form two simultaneous equations and solve them. (4 marks)

(c) Make a the subject of the formula $v = u + at$, and find a when $v = 30$, $u = 10$ and $t = 4$. (3 marks)

4.

An incomplete table of values for the function $y = 2x - 3$ is given below.

x	–1	0	1	2	3
y	–5	…	–1	…	3

(i) Fill in the blanks in the table. (1 marks)

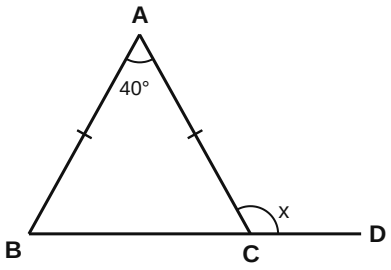
(ii) Using a suitable scale, draw the graph of the function on graph paper. (3 marks)

(iii) Write the gradient and the intercept of the graph. (2 marks)

(iv) Using the graph, find the value of x when $y = 0$. (2 marks)

(v) Write the equation of the line parallel to this graph which passes through the point (0, 2). (2 marks)

5.



(a) (i) Find the sum of the interior angles of a pentagon. (2 marks)

(ii) The interior angles of a pentagon are x, 2x, 110°, 120° and 130°. Find the value of x. (2 marks)

(iii) Each exterior angle of a regular polygon is 40°. Find the number of sides. (2 marks)

(b) In the figure, AB = AC and BÂC = 40°. The side BC is produced to D. Find the value of x, giving reasons. (4 marks)

6.

(π = 22/7)

(a) A ladder of length 13 m rests against a vertical wall. The foot of the ladder is 5 m from the wall on horizontal ground. Find how high up the wall the ladder reaches. (3 marks)

(b) The radius of a bicycle wheel is 35 cm.

(i) Find the circumference of the wheel. (2 marks)

(ii) How many complete turns does the wheel make in travelling 1.1 km? (2 marks)

(c) Find the perimeter of a semicircular lamina of diameter 14 cm. (3 marks)

7.

The marks (out of 10) obtained by 20 students for a quiz are shown in the frequency table below.

Marks	4	5	6	7	8
No. of students	2	5	6	4	3

(i) Write the mode of the marks. (1 marks)

(ii) Find the median of the marks. (2 marks)

(iii) Find the mean of the marks. (3 marks)

(iv) Find the range of the marks. (1 marks)

(v) If this information is shown in a pie chart, find the angle of the sector representing the students who obtained 6 marks. (3 marks)